MAKING TWO MAPS OF HYPERCOVERINGS HOMOTOPIC AFTER REFINEMENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2074-2083 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 2085-2122, the complete author proof (36 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_hypercovering.json (pb=1390, stmt 2074-2083, proof 2085-2122).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01GN: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1890–1904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01GH: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1511–1536. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01GP: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1996–2009. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let X be an object of $\mathcal{C}$. Let K, L be hypercoverings of X. Let $a, b : K \to L$ be morphisms of hypercoverings. There exists a morphism of hypercoverings $c : K' \to K$ such that $a \circ c$ is homotopic to $b \circ c$.*

Date: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Consider the following commutative diagram

$$\begin{array}{ccccc} K' & \overset{def}{=\!=} & K \times_{(L\times L)} \mathrm{Hom}(\Delta[1], L) & \longrightarrow & \mathrm{Hom}(\Delta[1], L) \\ & \searrow^{c} & \downarrow & & \downarrow \\ & & K & \xrightarrow{(a,b)} & L \times L \end{array}$$

By the functorial property of $\mathrm{Hom}(\Delta[1], L)$ the composition of the horizontal morphisms corresponds to a morphism $K' \times \Delta[1] \to L$ which defines a homotopy between $c \circ a$ and $c \circ b$. Thus if we can show that K' is a hypercovering of X, then we obtain the lemma. To see this we will apply Lemma 01GH to the pair of morphisms $K \to L \times L$ and $\mathrm{Hom}(\Delta[1], L) \to L \times L$. Condition (1) of Lemma 01GH is satisfied. Condition (2) of Lemma 01GH is true because $\mathrm{Hom}(\Delta[1], L)_0 = L_1$, and the morphism $(d_0^1, d_1^1) : L_1 \to L_0 \times L_0$ is a covering of $\mathrm{SR}(\mathcal{C}, X)$ by our assumption that L is a hypercovering. To prove condition (3) of Lemma 01GH we use Lemma 01GP above. According to this lemma the morphism γ of condition (3) of Lemma 01GH is the morphism

$$\mathrm{Hom}(\Delta[1] \times \Delta[n+1], L)_0 \longrightarrow \mathrm{Hom}(U, L)_0$$

where $U \subset \Delta[1] \times \Delta[n+1]$. According to Lemma 01GN this is a covering and hence the claim has been proven. □